

Descriptive Statistics and Frequentist Statistics

Module 3

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Module Outline

Descriptive Statistics

- Histogram

- Sample Mean and Variance

- Order Statistics

- Sample Covariance

- Sample Covariance Matrix

Frequentist Statistics

- Sampling

- Mean Square Error

- Consistency

- Confidence Intervals

- Parametric Model Estimation

- Non-parametric Model Estimation

Summary

What is Descriptive Statistics?

Definition. Descriptive statistics summarizes and organizes the features of a data set using numerical measures, tables, and graphs, without drawing conclusions that go beyond the observed data.

- ▶ Goal: describe **center**, **spread**, **shape**, and **relationships**.
- ▶ Graphical summaries: histogram, boxplot, scatterplot, stem-and-leaf plot, bar chart, pie chart.
- ▶ Numerical summaries: mean, median, mode, variance, standard deviation, order statistics, covariance, correlation.
- ▶ Descriptive statistics makes *no probabilistic claim* about a population; it only reports what is present in the sample. Contrast this with inferential (frequentist) statistics, where we use the sample to say something about a population we have not fully observed.

Histogram: Definition and Construction

Definition. A histogram is a graphical estimate of the distribution of a numerical variable, built by (1) partitioning the range of the data into contiguous bins, (2) counting the number of observations in each bin, and (3) drawing bars whose *area* is proportional to that count.

For equal-width bins of width h , the **density-scaled** height is

$$\hat{f}(x) = \frac{n_i}{nh}, \quad x \in [a_{i-1}, a_i)$$

so that the total area under the histogram is exactly 1, mimicking a probability density function.

Note. A frequency histogram (raw counts on the y -axis) and a density histogram (as above) have the same shape; only the vertical scale differs. Density scaling is what allows the histogram to be compared directly with a theoretical pdf overlay.

Histogram: Choice of Bin Width

Bin choice is a bias–variance trade-off:

- ▶ Too few bins (wide h) $\Rightarrow$ **oversmoothed**, hides real structure (multi-modality, skew) — high bias.
- ▶ Too many bins (narrow h) $\Rightarrow$ **noisy/spiky** — high variance, dominated by sampling noise.

Common automatic rules (n = sample size, s = sample std. dev., IQR = interquartile range):

$$\text{Sturges: } k = \lceil \log_2 n \rceil + 1 \qquad \text{Scott: } h = \frac{3.49 s}{n^{1/3}}$$

$$\text{Freedman–Diaconis: } h = \frac{2 \text{IQR}}{n^{1/3}}$$

Note. Freedman–Diaconis is the most robust of the three because IQR is insensitive to outliers, whereas Scott's rule (which uses s) can be badly distorted by even one extreme observation.

Histogram as a Non-Parametric Density Estimator

- ▶ The histogram is the simplest **non-parametric density estimator**: it estimates the unknown pdf $f(x)$ using only the data, with no assumed functional form for f .
- ▶ For a fixed bin $[a_{i-1}, a_i)$ with true probability $p_i = \int_{a_{i-1}}^{a_i} f(x) dx$, the count n_i is Binomial(n, p_i), so $\hat{f}(x) = n_i/(nh)$ is unbiased for p_i/h (an average of f over the bin, not exactly $f(x)$ at a point).
- ▶ **Consistency** of the histogram as an estimator of f requires both $h \rightarrow 0$ and $nh \rightarrow \infty$ as $n \rightarrow \infty$: bins must shrink (to reduce bias) but still contain enough points on average (to reduce variance).

Sample Mean

Let $X_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 .

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Properties:

- ▶ **Unbiased:** $\mathbb{E}[\bar{X}] = \mu$.
- ▶ **Variance:** $\text{Var}(\bar{X}) = \sigma^2/n$ — shrinks as n grows, so $\bar{X}$ becomes more precise with more data.
- ▶ **Law of Large Numbers (LLN):** $\bar{X} \xrightarrow{P} \mu$ as $n \rightarrow \infty$.
- ▶ **Central Limit Theorem (CLT):** $\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$, so for large n , $\bar{X} \approx N(\mu, \sigma^2/n)$ regardless of the shape of the original population distribution.

Sample Variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right)$$

- ▶ Dividing by $n - 1$ (**Bessel's correction**) makes S^2 **unbiased**: $\mathbb{E}[S^2] = \sigma^2$.
- ▶ The “plug-in” / MLE-style version divides by n instead:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2, \quad \mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2 \quad (\text{biased low})$$

- ▶ $n - 1 =$ **degrees of freedom** remaining after estimating μ with $\bar{X}$.

Why $n - 1$? The deviations satisfy $\sum_i (X_i - \bar{X}) = 0$, so only $n - 1$ of them are free to vary once $\bar{X}$ is fixed; using $\bar{X}$ (which is, on average, closer to each X_i than the unknown true μ is) makes $\sum (X_i - \bar{X})^2$ systematically smaller than $\sum (X_i - \mu)^2$, and dividing by the smaller number $n - 1$ exactly compensates.

Other Measures of Center and Spread

- ▶ **Median:** robust measure of center, unaffected by extreme outliers (unlike the mean).
- ▶ **Mode:** most frequent value; useful for categorical or multi-modal data.
- ▶ **Range:** $X_{(n)} - X_{(1)}$; highly sensitive to outliers.
- ▶ **Standard deviation:** $S = \sqrt{S^2}$, same units as the data (variance is in squared units).
- ▶ **Coefficient of variation:** $CV = S/\bar{X}$, a unit-free measure of relative spread, useful for comparing variability across data sets with different units or scales.
- ▶ **Skewness and kurtosis:** standardized third and fourth central moments, describing asymmetry and tail-heaviness of the distribution's shape.

Order Statistics: Definition

Arrange the sample $X_1, \dots, X_n$ in increasing order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

$X_{(k)}$ is the k -th **order statistic**.

- ▶ $X_{(1)} = \min$, $X_{(n)} = \max$, Range $R = X_{(n)} - X_{(1)}$.
- ▶ **Median:** $X_{((n+1)/2)}$ (n odd), $\frac{1}{2} (X_{(n/2)} + X_{(n/2+1)})$ (n even)
- ▶ Order statistics are **not independent** of each other (even if the original X_i are i.i.d.), because knowing $X_{(1)}$ constrains what $X_{(2)}, \dots$ can be.

Quantiles, Quartiles, IQR, Boxplot

- ▶ The **p -th sample quantile** is (approximately) the value below which a fraction p of the data falls; computed by interpolating between order statistics near rank $p(n - 1) + 1$ (conventions vary slightly across software).
- ▶ **Quartiles**: Q_1 (25th percentile), $Q_2 = \text{median}$ (50th), Q_3 (75th percentile).
- ▶ **Interquartile range**: $\text{IQR} = Q_3 - Q_1$ — spread of the middle 50% of data, robust to outliers.
- ▶ **Five-number summary**: $X_{(1)}, Q_1, \text{Median}, Q_3, X_{(n)}$ — basis of the boxplot.
- ▶ **Outlier rule of thumb**: flag points outside $[Q_1 - 1.5 \text{IQR}, Q_3 + 1.5 \text{IQR}]$ (Tukey's fences).

Sampling Distribution of Order Statistics

For a continuous r.v. with pdf f and cdf F , the density of the k -th order statistic (out of n i.i.d. draws) is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1-F(x)]^{n-k} f(x)$$

Special cases:

$$f_{X_{(1)}}(x) = n[1-F(x)]^{n-1}f(x), \quad f_{X_{(n)}}(x) = n[F(x)]^{n-1}f(x)$$

Example. For $X_i \sim \text{Uniform}(0, 1)$, $F(x) = x$, so $f_{X_{(n)}}(x) = nx^{n-1}$ on $(0, 1)$, giving $\mathbb{E}[X_{(n)}] = n/(n+1) \rightarrow 1$ as $n \rightarrow \infty$: the sample maximum approaches the true upper endpoint.

Sample Covariance

For paired data $(X_1, Y_1), \dots, (X_n, Y_n)$:

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n-1} \left(\sum_i X_i Y_i - n\bar{X}\bar{Y} \right)$$

- ▶ $S_{XY} > 0$: X and Y tend to increase/decrease together.
- ▶ $S_{XY} < 0$: X and Y tend to move in opposite directions.
- ▶ $S_{XY} \approx 0$: little or no *linear* association.
- ▶ Magnitude depends on the units of X and Y , so raw covariance values are **not comparable** across different variable pairs — this motivates the correlation coefficient.

Sample Correlation Coefficient

$$r = \frac{S_{XY}}{S_X S_Y} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2} \sqrt{\sum_i (Y_i - \bar{Y})^2}}$$

- ▶ $-1 \leq r \leq 1$; unit-free and scale-invariant.
- ▶ $|r| = 1 \Leftrightarrow$ the points lie exactly on a straight line.
- ▶ $r = 0$: no linear relationship (nonlinear relationship still possible).
- ▶ r is unchanged by any positive linear transform of X or Y (e.g. changing units from cm to inches), and flips sign under a negative linear transform.

Sample Covariance Matrix

For p -dimensional observations $\mathbf{X}_1, \dots, \mathbf{X}_n \in \mathbb{R}^p$ with sample mean vector $\bar{\mathbf{X}} = \frac{1}{n} \sum_i \mathbf{X}_i$:

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$$

In matrix form, with data matrix $\mathbf{X}$ ($n \times p$, rows = observations) and centering matrix $\mathbf{H} = \mathbf{I}_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$:

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{H} \mathbf{X}$$

- ▶ $\mathbf{S}$ is $p \times p$, **symmetric** and **positive semi-definite**.
- ▶ Diagonal entry S_{jj} = sample variance of variable j ; off-diagonal S_{jk} = sample covariance of variables j, k .

Sample Correlation Matrix

Let $\mathbf{D} = \text{diag}(\sqrt{S_{11}}, \dots, \sqrt{S_{pp}})$.

$$\mathbf{R} = \mathbf{D}^{-1}\mathbf{S}\mathbf{D}^{-1}$$

- ▶ $\mathbf{R}$ has 1's on the diagonal, entries in $[-1, 1]$ elsewhere.
- ▶ Standardizes each variable, removing the effect of differing units/scales, so that variables can be compared directly.
- ▶ Both $\mathbf{S}$ and $\mathbf{R}$ are the starting point for multivariate techniques:
 - ▶ **PCA**: eigen-decomposition of $\mathbf{S}$ (or $\mathbf{R}$) to find directions of maximal variance.
 - ▶ **Mahalanobis distance**: uses $\mathbf{S}^{-1}$ to measure distance accounting for correlation between variables.
 - ▶ Multivariate normal likelihood is parameterized directly by a covariance matrix.

Frequentist Statistics: Overview

Definition. Under the frequentist interpretation, probability is the long-run relative frequency of an event under (hypothetical) repeated sampling. Population parameters (θ) are treated as fixed, unknown constants; randomness enters only through the sampling process, not through θ itself.

- ▶ Contrast with Bayesian statistics, which treats θ itself as a random variable with a prior distribution.
- ▶ Core frequentist tools: point estimation, sampling distributions, bias/variance/MSE, consistency, confidence intervals, hypothesis testing.
- ▶ This module covers: sampling, MSE, consistency, confidence intervals, and parametric & non-parametric estimation.

Sampling: Basic Concepts

- ▶ **Population:** entire set of units of interest. **Sample:** the observed subset.
- ▶ **Random (i.i.d.) sample:** $X_1, \dots, X_n$ independent, each distributed as the population distribution F_θ .
- ▶ **Statistic:** a function $T(X_1, \dots, X_n)$ of the sample that does not depend on unknown parameters (e.g. $\bar{X}$, S^2 , sample max).
- ▶ **Sampling distribution:** the probability distribution of a statistic T , induced purely by the randomness in the sample.
- ▶ **Standard error:** the standard deviation of a statistic's sampling distribution, e.g. $SE(\bar{X}) = \sigma/\sqrt{n}$.

Sampling Methods

- ▶ **Simple random sampling (SRS)**: every subset of size n is equally likely to be chosen.
- ▶ **Stratified sampling**: divide the population into homogeneous strata, sample independently within each — reduces variance when strata differ substantially.
- ▶ **Cluster sampling**: divide into clusters, randomly select whole clusters — cheaper logistically, but usually less efficient than SRS since units within a cluster tend to be similar.
- ▶ **Systematic sampling**: choose every k -th unit from an ordered list, with a random starting point.
- ▶ **With vs. without replacement**: sampling with replacement preserves exact independence; sampling without replacement from a finite population of size N requires the **finite population correction** factor $\sqrt{(N - n)/(N - 1)}$ multiplying the standard error.

Sampling Distribution of the Mean

If $X_1, \dots, X_n$ i.i.d. with mean μ , variance σ^2 :

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

- ▶ Population exactly Normal $\Rightarrow \bar{X} \sim N(\mu, \sigma^2/n)$ **exactly**, for every n .
- ▶ Otherwise, by the CLT, for large n : $\bar{X} \approx N(\mu, \sigma^2/n)$ approximately.
- ▶ **Rule of thumb**: CLT approximation is usually adequate for $n \gtrsim 30$ for moderately skewed populations; more skewed or heavy-tailed populations need larger n .

Mean Square Error (MSE)

For an estimator $\hat{\theta}$ of parameter θ :

$$\text{MSE}(\hat{\theta}) = \mathbb{E} [(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2, \quad \text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

Derivation:

$$\begin{aligned} \mathbb{E}[(\hat{\theta} - \theta)^2] &= \mathbb{E} \left[((\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta))^2 \right] \\ &= \text{Var}(\hat{\theta}) + 2(\mathbb{E}[\hat{\theta}] - \theta) \underbrace{\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]]}_{=0} + (\mathbb{E}[\hat{\theta}] - \theta)^2 \\ &= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2 \end{aligned}$$

MSE: Interpretation, Trade-offs, Example

- ▶ MSE is the standard criterion to compare competing estimators — smaller MSE is better.
- ▶ Unbiased estimator: $\text{Bias} = 0 \Rightarrow \text{MSE} = \text{Var}$.
- ▶ A slightly **biased** estimator can have a much smaller variance, giving *lower total MSE* than an unbiased alternative — the bias–variance trade-off (e.g. ridge regression vs. ordinary least squares).

Example. Compare $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$ (biased, MLE) with $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ (unbiased). Under normality it can be shown that $\hat{\sigma}^2$, despite its bias, has *smaller* MSE than S^2 for estimating σ^2 — biasedness alone does not make an estimator worse overall.

Consistency of Estimators

$\hat{\theta}_n$ is a **consistent** estimator of θ if it converges in probability to θ :

$$\hat{\theta}_n \xrightarrow{P} \theta \iff \forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0$$

Sufficient condition (mean-square consistency): if $\text{Bias}(\hat{\theta}_n) \rightarrow 0$ and $\text{Var}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$, then $\text{MSE}(\hat{\theta}_n) \rightarrow 0$, which implies $\hat{\theta}_n \xrightarrow{P} \theta$ (by Chebyshev's inequality applied to $\hat{\theta}_n - \theta$).

- ▶ $\bar{X}$ is consistent for μ : $\text{Bias} = 0$, $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$.
- ▶ S^2 is consistent for σ^2 .
- ▶ $\hat{\sigma}^2$ (MLE, divide by n) is also consistent for σ^2 , even though biased for every finite n .

Consistency vs. Unbiasedness

- ▶ **Unbiasedness** is a *finite-sample* property: $\mathbb{E}[\hat{\theta}_n] = \theta$ must hold for every fixed n .
- ▶ **Consistency** is an *asymptotic* property: behavior only as $n \rightarrow \infty$.
- ▶ Possible combinations:
 - ▶ Unbiased *and* consistent — e.g. $\bar{X}$, S^2 .
 - ▶ Biased but consistent — e.g. MLE variance $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$: biased for finite n , but bias $\rightarrow 0$ as $n \rightarrow \infty$.
 - ▶ Unbiased but *not* consistent — possible in pathological/contrived constructions (e.g. an estimator that ignores most of the data and has non-vanishing variance).
 - ▶ Neither unbiased nor consistent — a poorly chosen estimator.

Confidence Intervals: Definition and Interpretation

A $100(1 - \alpha)\%$ confidence interval (CI) for θ is a random interval $[L(\mathbf{X}), U(\mathbf{X})]$ satisfying

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Correct frequentist interpretation: if the sampling procedure were repeated many times and a CI constructed each time, approximately $100(1 - \alpha)\%$ of those intervals would contain the true, fixed θ . The interval is random; θ is fixed.

Common wrong interpretation to avoid: “there is a 95% probability that θ lies in *this particular* computed interval” — once data are observed and L, U are specific numbers, θ either is or is not in $[L, U]$; there is no probability left in a frequentist framework.

CI for the Mean: σ Known

If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ (or n large via CLT) and σ is **known**:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the upper- $\alpha/2$ quantile of $N(0, 1)$ ($z_{0.025} = 1.96$ for a 95% CI; $z_{0.005} = 2.576$ for 99%).

Derivation: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ (this Z is a **pivotal quantity** — see later slide).

Then $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$; rearranging the inequality to isolate μ gives the interval above.

CI for the Mean: σ Unknown (t-interval)

Replace σ with sample standard deviation S ; use Student's t -distribution with $n - 1$ degrees of freedom:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

- ▶ $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ when the population is normal.
- ▶ t -distribution has heavier tails than $N(0, 1) \Rightarrow t_{\alpha/2, n-1} > z_{\alpha/2}$, reflecting the extra uncertainty from also having to estimate σ .
- ▶ As $n \rightarrow \infty$, $t_{n-1} \rightarrow N(0, 1)$, so the t - and z -intervals converge for large samples.

CI for a Proportion

For large n , with sample proportion $\hat{p} = X/n$ (X = number of successes out of n Bernoulli trials):

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

based on the normal approximation to the Binomial distribution, generally considered valid when $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$.

Note. An alternative, often more accurate for small/moderate n , is the **Wilson score interval**, which does not rely on the same crude normal approximation and performs better near $\hat{p}$ close to 0 or 1.

CI for Variance

If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

100(1 - α)% CI for σ^2 :

$$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right)$$

where $\chi_{\alpha/2, n-1}^2$ is the upper- $\alpha/2$ quantile of the chi-square distribution with $n - 1$ degrees of freedom.

General Construction: Pivotal Quantities

Definition. A pivotal quantity is a function $Q(\mathbf{X}, \theta)$ of the data and the parameter whose sampling distribution does not depend on θ or any other unknown parameter.

- ▶ **Recipe:** (1) find a pivot Q ; (2) using its known distribution, find a, b with $P(a \leq Q \leq b) = 1 - \alpha$; (3) algebraically invert the inequality to isolate θ .
- ▶ Pivots seen so far: $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$, $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$, $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$.
- ▶ CI width typically shrinks like $1/\sqrt{n}$ — more data gives a more precise (narrower) interval, all else equal.

Parametric Estimation: Setup

Definition. A parametric model assumes data come from a distribution family $\{f_\theta : \theta \in \Theta\}$ indexed by a finite-dimensional parameter $\theta \in \Theta \subseteq \mathbb{R}^k$ (e.g. $\text{Normal}(\mu, \sigma^2)$, $\text{Binomial}(n, p)$, $\text{Poisson}(\lambda)$, $\text{Exponential}(\lambda)$). Main estimation methods covered:

1. Method of Moments (MoM)
2. Maximum Likelihood Estimation (MLE)
3. (Brief mention) Least Squares — for regression-type models
4. (Brief mention) Bayesian estimation — for contrast; not strictly frequentist since it treats θ as random with a prior.

Method of Moments

Idea: equate population moments $\mu_k = \mathbb{E}[X^k]$ to sample moments $m_k = \frac{1}{n} \sum_i X_i^k$ and solve for θ .

Example. Normal(μ, σ^2):

$$\hat{\mu}_{MoM} = \bar{X}, \quad \hat{\sigma}_{MoM}^2 = \frac{1}{n} \sum_i (X_i - \bar{X})^2$$

Example. Exponential(λ), where $\mathbb{E}[X] = 1/\lambda$:

$$\hat{\lambda}_{MoM} = 1/\bar{X}$$

Example. Uniform($0, \theta$), where $\mathbb{E}[X] = \theta/2$:

$$\hat{\theta}_{MoM} = 2\bar{X}$$

Note. MoM estimators are simple to compute but can sometimes fall outside the valid parameter space (e.g. $2\bar{X}$ could in principle be less than $X_{(n)}$'s observed maximum) and are generally less statistically efficient than MLE.

Maximum Likelihood Estimation (MLE)

Likelihood function:

$$L(\theta \mid x_1, \dots, x_n) = \prod_{i=1}^n f_{\theta}(x_i)$$

MLE:

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} L(\theta) = \arg \max_{\theta} \ell(\theta), \quad \ell(\theta) = \log L(\theta)$$

found (when ℓ is differentiable) by solving the **score equation** $\partial \ell(\theta) / \partial \theta = 0$.

Properties (under regularity conditions), as $n \rightarrow \infty$:

- ▶ Consistent: $\hat{\theta}_{MLE} \xrightarrow{P} \theta$.
- ▶ Asymptotically normal and asymptotically efficient (attains the Cramér–Rao lower bound in the limit).
- ▶ **Invariance property:** the MLE of $g(\theta)$ is $g(\hat{\theta}_{MLE})$ for any function g .

MLE Example: Normal Distribution

For $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Setting $\partial\ell/\partial\mu = 0$ and $\partial\ell/\partial\sigma^2 = 0$ and solving jointly:

$$\hat{\mu}_{MLE} = \bar{X}, \quad \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2$$

Note $\hat{\sigma}_{MLE}^2$ is **biased** (divides by n , not $n - 1$) but is still **consistent**.

MLE Example: Bernoulli/Binomial and Poisson

Example. Bernoulli(p), data $X_1, \dots, X_n \in \{0, 1\}$:

$$\ell(p) = \left(\sum_i x_i \right) \log p + \left(n - \sum_i x_i \right) \log(1 - p) \Rightarrow \hat{p}_{MLE} = \frac{1}{n} \sum_i x_i = \bar{X}$$

Example. Poisson(λ):

$$\ell(\lambda) = -n\lambda + \left(\sum_i x_i \right) \log \lambda - \sum_i \log(x_i!) \Rightarrow \hat{\lambda}_{MLE} = \bar{X}$$

- In both cases the MLE coincides exactly with the sample mean and with the Method-of-Moments estimator — this is common (though not universal) for simple one-parameter exponential-family models.

Fisher Information and the Cramér–Rao Lower Bound

Fisher information (single observation):

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial \log f_{\theta}(X)}{\partial \theta} \right)^2 \right] = - \mathbb{E} \left[\frac{\partial^2 \log f_{\theta}(X)}{\partial \theta^2} \right]$$

Cramér–Rao lower bound (CRLB): for any unbiased estimator $\hat{\theta}$ based on n i.i.d. observations,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n I(\theta)}$$

An estimator achieving this bound with equality is called **efficient**. MLE is asymptotically efficient (achieves the bound as $n \rightarrow \infty$, under regularity conditions).

Non-parametric Estimation: Setup

Definition. A non-parametric model makes no finite-dimensional assumption about the form of f or F ; the object being estimated (the whole function) is effectively infinite-dimensional. **Trade-off vs. parametric methods:**

- ▶ **Parametric:** highly efficient *if* the assumed model is correct; can be badly biased if the model is wrong (misspecified).
- ▶ **Non-parametric:** robust to misspecification (fewer assumptions to get wrong); typically requires more data and converges more slowly than a correctly specified parametric model would.

Empirical Distribution Function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}$$

- ▶ Natural non-parametric estimator of the true cdf F : a step function jumping by $1/n$ at each observed data point.
- ▶ For fixed x : $n\hat{F}_n(x) \sim \text{Binomial}(n, F(x))$, so $\hat{F}_n(x)$ is **unbiased** with $\text{Var}(\hat{F}_n(x)) = F(x)(1 - F(x))/n$.
- ▶ **Glivenko–Cantelli theorem** (“fundamental theorem of statistics”):

$$\sup_x |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0 \quad \text{as } n \rightarrow \infty$$

i.e. uniform, almost-sure convergence of the entire estimated curve to the true cdf, not just pointwise convergence.

Kernel Density Estimation (KDE)

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

- ▶ $K(\cdot)$: kernel function (e.g. Gaussian, Epanechnikov), typically symmetric and integrating to 1.
- ▶ $h > 0$: **bandwidth**, controlling smoothness — the continuous analogue of histogram bin width.
- ▶ **Bias–variance trade-off in h** : small h gives low bias but high variance (wiggly estimate); large h gives high bias (oversmoothed) but low variance.
- ▶ KDE generalizes the histogram by replacing the rigid, hard-edged bin-membership indicator with a smooth, weighted kernel — this removes the artificial discontinuities (jumps) present in a histogram.

Non-parametric Regression

Nadaraya–Watson kernel regression estimator:

$$\hat{m}(x) = \frac{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{x-X_i}{h}\right)}$$

- ▶ A kernel-weighted local average of the Y_i near x , estimating $m(x) = \mathbb{E}[Y \mid X = x]$ without assuming any specific functional form (e.g. not assuming linearity as ordinary linear regression does).
- ▶ Related non-parametric regression methods: local polynomial regression, smoothing splines, k -nearest-neighbours regression.
- ▶ **Convergence rates:** non-parametric estimators typically converge more slowly than correctly specified parametric estimators — e.g. KDE/Nadaraya–Watson rate is roughly $n^{-2/5}$, versus $n^{-1/2}$ for a correctly specified parametric MLE — the statistical price paid for not assuming a model form.

Bootstrap: A Non-parametric Resampling Method

- ▶ **Idea:** treat the observed empirical distribution $\hat{F}_n$ as a stand-in for the true (unknown) F , and simulate the sampling distribution of a statistic by repeatedly resampling *with replacement* from the observed data.
- ▶ **Procedure:** draw B resamples of size n from $X_1, \dots, X_n$ (with replacement); compute the statistic of interest $\hat{\theta}_b^*$ on each resample; use the empirical distribution of $\{\hat{\theta}_1^*, \dots, \hat{\theta}_B^*\}$ to approximate the sampling distribution of $\hat{\theta}$, estimate its standard error, or build a confidence interval.
- ▶ Useful when the sampling distribution of a statistic has no simple closed-form expression (e.g. for the sample median or a complicated function of parameters).

Module Summary

Descriptive Statistics

- ▶ Histogram (bin-width rules, density estimate)
- ▶ Sample mean $\bar{X}$, sample variance S^2
- ▶ Order statistics, quantiles, IQR, boxplot
- ▶ Sample covariance, correlation
- ▶ Sample covariance/correlation matrix, link to PCA

Frequentist Statistics

- ▶ Sampling, sampling distributions, standard error
- ▶ $\text{MSE} = \text{Variance} + \text{Bias}^2$
- ▶ Consistency vs. unbiasedness
- ▶ Confidence intervals via pivotal quantities
- ▶ Parametric estimation: MoM, MLE, Fisher info, CRLB
- ▶ Non-parametric: EDF, Glivenko–Cantelli, KDE, Nadaraya–Watson, bootstrap

Thank You